CPD Implementation Audit: Ranked Issues and Repair Boundaries

7 November 2026

1 Scope, evidence and interpretation

This is a static audit of revision `c9d46ecbaea3500ee5cf1d3b8b278a9d780dd0d1`. Line numbers refer to that checkout. No project code, tests, simulations, data processing was executed. The document was subsequently compiled at the user's request; compilation is not validation of CPD behavior. No pipeline code was modified.

The initial audit used the pipeline-level explanation in `docs/pipeline.tex`: patient-specific unsupervised adaptation may use a held-out patient's available past; clinical classifiers train on training patients; onset features must not use future readings; gaps are defined on actual observations; comparisons must have explicit feature bundles. Dedicated historical CPD notes are explanatory context, not a requirement to restore an obsolete algorithm. Commit `8095fe3` deliberately replaced whole-sequence HMM EM/Viterbi with fixed-parameter forward inference and introduced the present whitening/calibration design.

PDF verification, 7 November 2026. The actual `docs/pipeline.pdf` was subsequently extracted and checked directly. It is the exported *Asthma Prediction Protocol / Interactive Asthma Prediction Report*, not a compiled copy of `pipeline.tex`. Its PDF metadata identifies Chrome/Skia as the producer and a creation date of 4 November 2026. It contains two very tall PDF pages; the section numbers below are more useful than page numbers. The PDF is the system-design reference for the alignment table that follows.

2 Explicit alignment with the system pipeline PDF

PDF section	Required system behavior	Relevant audit findings
1: objectives	Detection asks about day D; onset asks about the following 1–3 days. Features include day D, not subsequent observations.	C01, C05, C06, C11, C19 address information availability. Full-day features themselves are allowed.
2: available measurements; 4: invariant variables	Every path gives the classifier all specified measurements. The processing path changes HR/HRV features; temp, steps and sleep remain available raw.	C03 is a direct implementation-contract failure. C18 identifies branch-dependent support of nominally raw drift inputs.

PDF section	Required system behavior	Relevant audit findings
3: signal processing; 3.7: phases	Compute each channel's derived signals once, store them, then assemble paths without repeating work per classifier/fold.	C04 requires correct content dependencies, not moving all CPD fitting into every fold. Patient-local past-only adaptation is compatible with shared cached artifacts.
3.1; 4: path matrix and path 5 flow	Smoothing supplies raw, baseline and residual; CPD reads the path's designated signal and emits features.	C03, C18 and C24 check preservation, support and aggregation. Nightly raw-level drift is an added implementation branch whose role should be explicit in each comparison.
3.2/3.5; A.2: lookahead	Whole-record smoother calibration is recognized as retrospective; real-time settings must use past data.	C06 is a conditional real-time limitation already acknowledged by the PDF, not a demand to discard retrospective smoother experiments. C01 and C05 are additional CPD-local leaks.
4: invariant patients and split	Keep all 191 patients under the stated default eligibility; unavailable method outputs are blank, not removed days. Use the same patient split across paths.	C28 repairs must preserve blank-feature rows and eligibility. Successful-only cohort restriction is not an acceptable repair.
4: tasks, windows and horizons	Keep tasks separate; window changes add history; horizon changes narrow labels without moving the denominator.	This audit proposes no target, window or eligibility change. CPD validity must not silently select different scoreable days.
4/5: scoring	Use consistent day, episode and probability-band evaluation.	C09, C14–C16, C20, C24 and C29 concern CPD event/exposure definitions and their features/reports. CPD alarms are not clinical episode labels or probabilities.
A.3: gap rule	A break of at least 180 minutes starts a new segment. Channels retain their own time grids.	C02 and C08 directly concern segmentation; C07, C13, C19, C25 and C32 concern how detector memory/output respects those boundaries. A common grid across channels is not proposed.
A.4: architectural rules	Reuse keyed computations and preserve manifests, modularity and controlled comparisons.	C04 and C31 strengthen provenance; targeted repairs should preserve stage separation and method discovery.

What the PDF does not specify. It does not provide a complete mathematical contract for whitening, null generation, HMM fitting, CUSUM recovery, analytic drift calibration or every alarm lifecycle. Its references to four detectors also lag the five-method implementation. Therefore R-labelled findings below are implementation assumptions requiring validation or a design decision, not invented PDF requirements. The five-method code is audited on its own stated behavior, without requiring a return to an older four-detector design. Proposed streaming interfaces are optional repair architecture, not a condition imposed by the PDF.

Alignment is not certification. This mapping establishes that the audit targets the intended

system. It does not establish correct detector mathematics, useful clinical signal, or an acceptable false-alarm/delay trade-off. Those require the regression and independent null/injection checks below, followed by the PDF's held-out predictive evaluation on unchanged patients, days and feature bundles.

3 Evidence and severity conventions

Evidence labels. D = a control-flow, data-flow or algebraic defect visible in source, with an explicit triggering condition; no claim about its observed frequency. R = a statistical/model-design risk requiring validation or a policy decision, rather than a proven coding error. V = a verification or observability gap. Some findings carry more than one label. A proposed fixture is a specification for later verification, not a test result. No finite static review guarantees that all bugs have been found, and this report does not estimate clinical harm or metric inflation from unavailable data.

How to use the categories. The confirmed-defect section is the implementation repair list. The design/validation section is a decision and evidence agenda, and the verification section is an assurance checklist. Mixed findings are conservatively retained in the decision section rather than promoted wholesale to confirmed defects. All original locations, conditions, consequences, repair options and acceptance checks are retained.

Severity ranks by potential damage to conclusions and processing. S1 (critical) can invalidate causal claims or silently change/reuse an experimental input. S2 (major) can systematically corrupt a detector, null, alarm timing or calibration. S3 (moderate) distorts interpretation or affects particular edge cases. S4 (minor) affects robustness, reporting or maintenance. These are engineering priorities, not measured clinical severity. Within a tier, fix order depends on the branches actually used.

4 Register by evidence category and severity

Issue IDs are unchanged. Each category is sorted from S1 to S4; severity does not convert an unvalidated concern into a confirmed defect.

ID	Rank	Evidence	Issue
Confirmed implementation defects			
C01	S1	D	Shuffle calibration reads beyond its activation timestamp
C02	S1	D	Raw filler rows hide observation gaps
C03	S1	D	Channel metadata and feature-coverage bookkeeping are wrong
C04	S1	D	CPD cache omits shared code and side-input contents
C05	S1	D	Future sleep availability changes earlier drift windows
C07	S2	D	Held HR output crosses resets and settling boundaries
C08	S2	D	Shuffle null replaces original chunks with calendar days
C10	S2	D	Missing scores can become alarms at nonpositive thresholds
C11	S2	D	KCPD backdates scheduled results and revises earlier outputs
C14	S2	D	Consecutive nightly drift alarms are suppressed
C15	S2	D	Drift loses/delays nights and collapses simultaneous verdicts
C17	S2	D/V	STS accepts unreliable fits and has incomplete fallback handling
C32	S2	D	Long nonwear resumes with an empty baseline but remains armed
C19	S3	D	Partial-bin closure depends on later input or end-of-file
C24	S3	D	Daily event counts and time labels disagree with CPD accounting
C27	S3	D/V	Invalid settings and partially malformed inputs are not rejected
C28	S3	D/V	A patient error can abort a cohort and leave a running artifact
C29	S3	D/V	Calibration reports can compare different inputs and exposure sets
C33	S3	D/V	Missing one state source silently discards the other
Design decisions and empirical validation			

ID	Rank	Evidence	Issue
C06	S1	D/R	Upstream lookahead remains in CPD inputs
C09	S2	D/R	Calibration measures events but permits nearly continuous alarms
C12	S2	D/R	Whitening carries latent states across changed coordinates
C13	S2	D/R	Kalman CUSUM carries evidence indefinitely across gaps
C16	S2	R	Drift's analytic budget does not calibrate its actual process
C18	S2	D/R	STS null support is conditioned on upstream output availability
C20	S3	D/R	Worn-time and sleep-coverage accounting use nominal counts
C21	S3	D/R	Tail fitting discards scale; HMM uses a fixed scale thereafter
C22	S3	D/R	Calibration exclusions/fallbacks can include abnormal or old history
C23	S3	D/R	Severity values are not comparable across detectors
C25	S3	D/R	BOCPD gap semantics, phenotype and pruning need explicit bounds
C26	S3	R	STS synthetic fidelity and innovation-tail assignment are unverified
Verification and observability gaps			
C30	S2	V	Existing checks do not establish the advertised guarantees
C31	S4	D/V	Provenance and output omit important diagnostic state

5 What is trained, fitted, updated or fixed

Component	Current adaptation	Audit interpretation
BOCPD	Normal–Gamma posterior per run hypothesis; fixed prior and hazard	Online Bayesian updates are learning; no batch fit is missing.
KCPD	Window-specific bandwidth and permutation mean/SD	Local statistical estimation, not supervised training.
Kalman monitoring	Spike, cumulative sums, trailing pair statistic	Its fitted state-space noise model is upstream.
HMM	Forward probabilities; fixed dwell times and variance ratio	No EM by design; tail degree of freedom comes from patient burn-in.
Nightly drift	Rolling median/MAD and CUSUM	A fitted personal baseline plus an analytic threshold.
Whitening	Hour/state profiles, OU parameters, initial tail fit	Fitted on patient history at arming and later refits.
Shuffle	Thresholds fitted on resampled patient history	Shared unsupervised calibration affects all simulated streams.
STS	Gaussian state-space likelihood fit, separate tail fit, simulated thresholds	Genuine batch fitting at each calibration; must use past data only.

None of these CPD components trains on exacerbation labels. This is consistent with the pipeline. Whether HMM dwell times/emission parameters should instead be learned is a model-selection question. Reintroducing full-record EM or Viterbi would restore lookahead rather than repair it. Any supervised choice of detector settings must respect training/validation patient boundaries.

6 Confirmed implementation defects

These 19 findings identify a concrete source-visible implementation or contract defect. Confirmation here means a demonstrated code path or counterexample, not an executed reproduction or measured clinical impact. Fix the specified defect and run its acceptance check. The exact repair can still require an implementation choice; it must preserve the pipeline contract.

6.1 C01: Shuffle calibration exceeds its historical cutoff

Severity: S1. Evidence: D.

Location: `wearable/steps/detect.py`, lines 222–237.

Location: `wearable/cpd/_whiten.py`, lines 316–345.

`_thresholds` selects dates from indices before `p0`, but passes full-length arrays to `twin`. Its pools select *all* rows on each selected date. Arming midway through a day therefore includes later readings on that date. The ordinary later recalibration points are first-of-day indices, so their current date is not normally present in the prefix: the direct future-in-day counterexample is initial mid-day arming, not every weekly recalibration. Even without future rows, selecting a boundary date can include observations before the trailing-window lower bound. STS's direct slice is timestamp-bounded; its shuffle fallback inherits the shuffle defect. This can affect thresholds applied before their source data existed.

Targeted repair: Pass an explicitly bounded history object, with `stamp < t_cal` and the intended lower bound, into every null generator. Dates must never be a substitute for timestamp bounds.

Acceptance check (not run): Arm at 12:00, change only observations after 12:00, and require identical thresholds and earlier outputs. Repeat with STS failure forcing shuffle fallback.

6.2 C02: Raw-path segmentation is computed before removing fillers

Severity: S1. Evidence: D.

Location: `wearable/steps/detect.py`, lines 86–90 and 286–297.

Location: `wearable/steps/ingest.py`, lines 60–76.

Ingest inserts NaN filler rows every ten minutes across long gaps. Detection assigns chunks on adjacent *rows*, then removes nonfinite signal rows. Thus a four-hour observation gap becomes a series of ten-minute row gaps and no H8 boundary. Whitening misses its hard reset; KCPD's no-cross-boundary rule has no correct IDs to use. Exact 180-minute gaps can split while longer, filled gaps do not. Existing partially populated chunk columns are also accepted without validating every observed row. The separate chunk implementation defeats the shared segmentation contract.

Targeted repair: Construct/validate chunk IDs on the relevant actual observations with `wearable.chunking.assign_chunks`; map them to presentation rows afterwards. Specify whether missing transformed outputs retain raw-signal chunks.

Acceptance check (not run): Identical observed series with and without NaN fillers must produce identical observed-row chunks and detections, including 179/180/181-minute gaps.

6.3 C03: Artifact coverage changes the available model variables

Severity: S1. Evidence: D.

Location: `wearable/steps/detect.py`, lines 335–344 and 513–517.

Location: `wearable/steps/join.py`, lines 195–230.

Detection does not record `stats.signal`; join defaults to HRV when deciding which channel's raw block is covered. An HR-only detector can suppress raw HRV and duplicate raw HR. With the default two raw+CPD channels, HR is duplicated even when HRV happens to be present in its own detector table. Separately, the detected table retains only the chosen input column, not necessarily raw measurements. Merely adding `stats.signal` will therefore not repair all cases: a standalone residual, smoothed, surprise or tail detection artifact can suppress a raw block it does not contain. Standard experiment routes also pass smoother/forecast artifacts, which mitigates this second case

but not the incorrect detector metadata.

Targeted repair: Declare both signal identity and actual raw-feature coverage. Prefer joining a canonical raw block independently, with transformed features explicitly additive. Do not infer contents from channel identity alone.

Acceptance check (not run): Enumerate actual feature columns for HR-only, HRV-only and two-channel raw+CPD, residual+CPD and surprise+CPD, both standalone and through experiment recipes. All intended raw variables must appear once, on the same day support.

6.4 C04: CPD computation and dependencies are under-keyed

Severity: S1. Evidence: D.

Location: `wearable/steps/detect.py`, lines 259–267 and 455–470.

Location: `wearable/store.py`, lines 132–140.

Only detector-module hashes enter the CPD code hash. Changes to `steps/detect.py`, `_whiten.py`, `_sts.py`, or shared segmentation can reuse old detections. Raw input is keyed by the directory string rather than its contents; sleep/activity side files are read outside the declared upstream artifact. Their addition, correction or replacement changes results without changing the key. Recorded Git revision is provenance, not invalidation. Existing downstream artifacts can consequently survive a scientifically material CPD correction.

Targeted repair: Version/hash the complete CPD implementation and content-address the actual channel and state inputs. Introduce new logic keys when fixing these issues; rebuild dependent joins/models/evaluations with a documented invalidation boundary.

Acceptance check (not run): Changing each shared helper or one state-file value must change the appropriate CPD key; unchanged content must reuse it. No raw data need be committed.

6.5 C05: Future sleep flags select the definition of past nights

Severity: S1. Evidence: D.

Location: `wearable/cpd/drift.py`, lines 71–83.

`has_sleep` is computed across the entire supplied state array. A prefix with no sleep uses 00:00–06:00 raw readings. Appending the first sleep flag switches all earlier nights to sleep-only readings in the 18:00–12:00 window, potentially deleting those nights and changing baseline, alarms and magnitude. This is a CPD-internal causality failure, independent of smoother lookahead.

Targeted repair: Make availability/fallback decisions at each night's decision time using only state observations available then, with a declared minimum coverage policy.

Acceptance check (not run): Append the first sleep observation several days after arming; all previous verdicts and their baselines must remain unchanged.

6.6 C07: HR output holding leaks old flags across a reset

Severity: S2. Evidence: D.

Location: `wearable/cpd/_whiten.py`, lines 302–313.

Location: `wearable/steps/detect.py`, lines 324–359.

The holder maps a reading to the last bin's final-row index, regardless of chunk or elapsed gap. The first readings after a four-hour gap, before their new bin's last row, therefore hold the *previous chunk's* output, including its armed flag. An old alarm can appear during the new settling interval.

A shorter re-entry has the same issue. At midnight, old held outputs can also be attributed to the new day. This is not fixed by correctly segmenting the binned series.

Targeted repair: Apply current-reading chunk/re-entry/settling eligibility after mapping, and allow holding only within an explicit valid-time interval. Never borrow an armed state from another chunk. Keep verdict impulses distinct from held states.

Acceptance check (not run): End a chunk armed and flagged, then resume after a long gap with at least two observations in the first bin; no resumed reading may inherit the old alarm.

6.7 C08: Shuffle null destroys internal resets and adds artificial ones

Severity: S2. Evidence: D.

Location: `wearable/cpd/_whiten.py`, lines 328–345.

The supplied `chunk` array is unused: every sampled calendar day becomes one chunk. Within-day H8 boundaries vanish; midnight becomes an H8 boundary even for continuous wear. Noise filtering, settle exposure and KCPD validity thus differ between null and real data. Repeating days also introduces joins and level transitions that are not evidence of a physiological change in the original data.

Targeted repair: Preserve internal chunk boundaries and explicitly define/reset at synthetic block joins. Calibrate with the same reset and availability rules as real inference; document which original continuity is intentionally destroyed.

Acceptance check (not run): Map sampled days back to original rows and verify every internal reset, all synthetic joins, settling masks and eligible KCPD windows.

6.8 C10: Unavailable statistics become alarms at zero/negative thresholds

Severity: S2. Evidence: D.

Location: `wearable/cpd/kcpd.py`, lines 123–129.

Location: `wearable/cpd/kalman.py`, lines 85–91.

KCPD and Kalman convert NaN scores to zero before thresholding. Calibration retains missingness as non-hits, so inference has a different rule. If the learned threshold is zero or negative, invalid KCPD windows can alarm. Constant zero innovations give zero MMD scores; C09 can accept a zero threshold, so insufficient-coverage intervals then become positive too. Dividing severity by zero also yields NaN/inf. Kalman has the analogous problem for zero-valued streams. Outer arming does not establish that each individual detector statistic is available.

Targeted repair: Require finite scores in every hit mask. Represent unavailable separately from normal; make severity defined for all allowed thresholds.

Acceptance check (not run): A missing statistic can never alarm for any valid learned threshold; exercise zero/negative thresholds and degenerate nulls explicitly.

6.9 C11: KCPD backdates evaluations and can overwrite earlier readings

Severity: S2. Evidence: D.

Location: `wearable/cpd/kcpd.py`, lines 79–111.

For evaluation time T, the code writes to the last reading at or before T. When T lies between observations, that reading precedes availability of the scheduled result. Several grid evaluations in a gap can map to the same index; `last_val` is replaced and later written back onto it. A

longer recording extends the evaluation grid, potentially changing an earlier reading without a new observation near it. The arithmetic uses data no later than T, but the output's timestamp is still wrong.

Targeted repair: Keep evaluation timestamps as first-class output, or evaluate at actual arrival times and publish only at/after availability. Do not map a later clock event backwards. Reset/expire held statistics at declared gaps.

Acceptance check (not run): Use jittered arrivals, a long silent interval and many cuts inside that interval; already-published scores must be immutable.

6.10 C14: Consecutive nightly alarms collapse into one

Severity: S2. Evidence: D.

Location: `wearable/cpd/drift.py`, lines 126–137 and 148–154.

The CUSUM restarts after a verdict, but its above-threshold score remains held. If the next night independently alarms, the held Boolean never falls; rising-edge extraction deletes that second verdict. This contradicts once-per-verdict output.

Targeted repair: Return explicit verdict indices/IDs and one impulse for each alarmed night; keep held magnitude separate from event identity.

Acceptance check (not run): Two and three consecutive alarmed nights must produce two and three distinct verdict events, even with no below-threshold score between them.

6.11 C15: Night enumeration and delayed verdict mapping lose information

Severity: S2. Evidence: D.

Location: `wearable/cpd/drift.py`, lines 76–91 and 112–137.

Only calendar dates containing readings are enumerated. An evening's recorded sleep can yield a valid window ending on a date with no readings, but that night is never examined. Other nights are delayed to the first armed reading, potentially days later. Several nights can map to one index, where the held-score writer overwrites earlier verdicts. Their CUSUM/baseline updates still happened internally, so emitted output no longer describes the sequence processed. No maximum verdict delay exists.

Targeted repair: Generate candidate night-end dates from the window support; keep separate night time, availability time and publication time. Expire stale verdicts or retain them as explicit events rather than silently overwriting one row.

Acceptance check (not run): Recorded evening sleep followed by a no-reading next day; multiple nights completing before the next armed reading; recovery after extended nonwear.

6.12 C17: STS fit health is unchecked and fallback is incomplete

Severity: S2. Evidence: D/V.

Location: `wearable/cpd/_sts.py`, lines 126–139 and 178–191.

Location: `wearable/steps/detect.py`, lines 155–187.

The likelihood fit suppresses warnings and accepts returned parameters without checking optimizer convergence, finite values or useful variance estimates. The exception guard covers the main fit but not the later Student-t fit or simulation. Those failures escape instead of returning the documented shuffle fallback. A finite but unusable fit can generate degenerate nulls and bad thresholds. The

shared initial tail fit in `_whiten.py:221-224` is likewise unguarded.

Targeted repair: Validate fits and simulations explicitly; guard the whole calibration operation with narrowly classified errors; record fallback reason and fit diagnostics. Do not silently convert a numerical failure into a normal detector result.

Acceptance check (not run): Nonconverged optimizer, constant series, near-zero variances, invalid tail fits and simulation exceptions must produce declared fallback/failure status.

6.13 C32: A long absence leaves an armed detector with no fitted baseline

Severity: S2. Evidence: D.

Location: `wearable/cpd/_whiten.py`, lines 194–217 and 235–236.

Location: `wearable/cpd/_whiten.py`, lines 64–69 and 126–127.

Once arming has occurred it is never revoked. After an absence longer than `baseline_days`, the historical slice at the next refit can be empty. The fewer-than-ten fallback selects the same empty slice; it does not recover a valid baseline. Robust fitting then returns center 0 and scale 1, with no OU correlation. For a raw HR input near 80, this yields innovations near 80 rather than a meaningful patient-standardized score. The patient becomes eligible again after only the short settling interval, despite having no current fitted baseline. Earlier detector memory remains in use. Threshold recalibration can resample arbitrarily old history through C22. This differs from retaining state across a small gap.

Targeted repair: Require sufficient eligible fitting history after each material absence. Freeze an explicitly valid historical fit with a staleness policy, or re-enter calibration/unarmed status; never substitute generic center/scale while reporting ordinary armed inference.

Acceptance check (not run): An armed raw-channel patient disappears for more than 14 days and returns with unchanged physiology. Status and parameters must follow the declared re-entry policy and must not produce default-zero-centered pseudo-anomalies.

6.14 C19: A bin's closure time is inferred retrospectively

Severity: S3. Evidence: D.

Location: `wearable/cpd/_whiten.py`, lines 302–313.

Location: `wearable/cpd/kcpd.py`, lines 50–59.

Bins are timestamped at their final observed row, including the last incomplete bin at end-of-file. That row is not known to be the last until the bin closes or a later arrival establishes it. Truncating within a bin publishes a different partial median than extending the input. Even a completed sparse bin's last arrival can precede its known clock closure. This is an availability-time defect, not a centered future median at every row. The selftest explicitly excludes the final bin near cuts.

Targeted repair: Emit only closed bins at a declared clock boundary, or label partial outputs as provisional and exclude them from causal claims. Apply the same rule to KCPD's optional second binning layer.

Acceptance check (not run): Prefix cuts at every offset in a five-minute bin, including sparse bins and a closure with no new observation.

6.15 C24: Event and duration definitions differ across pipeline stages

Severity: S3. Evidence: D.

Location: `wearable/steps/detect.py`, lines 501–517.

Location: `wearable/steps/join.py`, lines 166–172.

Location: `wearable/cpd/_whiten.py`, lines 285–288.

Detection counts runs after deleting unarmed readings, intentionally merging across settling windows; it also merges across long gaps if flags remain high. Join counts on the full rows separately within each day: filler/unarmed zeros split events and midnight starts another daily bout. Columns ending in `_minutes` sum flagged rows rather than durations. HR bin holding expands a one-bin verdict onto several rows, so drift is no longer a single-row impulse on the original grid. These are different estimands unless explicitly named.

Targeted repair: Define event IDs and exposure/duration once, preserve them through mapping, and choose whether a daily count means new event onsets or events active that day. Rename sample counts or implement a documented bounded duration calculation.

Acceptance check (not run): One event across midnight, a gap and a settle interval; check global counts, daily counts, impulses and durations against the declared definitions.

6.16 C27: Input and setting validation does not protect numerical invariants

Severity: S3. Evidence: D/V.

Location: `wearable/steps/detect.py`, lines 64–90 and 443–475.

Location: `wearable/cpd/drift.py`, lines 55–59.

Settings names are checked, not numerical domains. Empty/duplicate method lists, nonpositive/nonfinite budgets, invalid dwell times, zero stride, zero permutations, nonpositive hazard scale, invalid direction or impossible coverage can crash or silently misbehave. A high requested analytic drift rate may put the root outside the fixed bracket; `cusum_k=0` divides by zero. Non-HRV channels without a CPD settings block inherit HRV cadence. Existing chunk columns with some NaNs are trusted; timestamps, strict ordering after duplicates, infinity, and array lengths are not validated at the public detector interface. A detector lacking both calibration hooks has no threshold entry when the pipeline invokes it.

Targeted repair: Validate domains, unique methods, channel support, timestamps, finite raw/transform alignment and detector calibration capabilities before creating an artifact. Reject unsupported cases with actionable errors, or implement declared fallbacks. Do not infer that ingestibility means CPD settings are appropriate.

Acceptance check (not run): A compact invalid-input matrix covering each listed invariant, including drift-only, an empty ensemble and a newly discovered detector with no score hook.

6.17 C28: One calibration failure can abort the entire cohort

Severity: S3. Evidence: D/V.

Location: `wearable/steps/detect.py`, lines 398–408 and 476–495.

`_patient_job` does not catch per-patient failures; materializing the process map propagates the exception and discards the usable returned results for writing. Only the all-empty-output case marks the artifact failed. Other exceptions can leave its manifest marked running. Global warning suppression hides numerical diagnostics. Conversely an unarmable patient with at least two readings returns a blank output and counts as a processed patient, without an explicit unarmable status.

Targeted repair: Preserve patient-level success/failure/unarmable reasons and a declared cohort completeness policy; mark the artifact failed on any unhandled exception. Scope warnings narrowly and retain diagnostics. Expose worker count rather than always creating almost one worker per

CPU for expensive STS tasks.

Acceptance check (not run): One patient fails while others succeed; verify manifest status, failure table, retained outputs, and unchanged modelling population/missingness policy.

6.18 C29: Calibration reports can compare different computations

Severity: S3. Evidence: D/V.

Location: `scripts/cpd_calibration_compare.py`, lines 35–47 and 55–87.

The "latest" selection filters ensemble, channel suffix, budget and calibration, but not `on`, smoother, input artifact identity, code version or cohort/support. It can compare STS residual detection with a newer raw shuffle run. The reader hardcodes CSV despite configurable artifact format. Exposure is computed from original armed rows and nominal cadence, whereas thresholds used binned support (C20). Labels such as "time flagged" are row fractions, not guaranteed elapsed-time fractions. Reported differences cannot all be attributed to calibration choice.

Targeted repair: Require a matched comparison specification and use the artifact table reader, keyed input identity and stored exposure definition. Print mismatches rather than selecting an arbitrary newest run.

Acceptance check (not run): Multiple raw/residual runs at the same budget, different smoothers and CSV/parquet artifacts; only truly matched pairs may be compared.

6.19 C33: State inputs use an all-or-nothing availability rule

Severity: S3. Evidence: D/V.

Location: `wearable/steps/detect.py`, lines 259–267 and 298–302.

If either sleep or steps file is absent, `_flags` returns None and discards the other source too. Available sleep flags therefore do not support nightly sleep windows when activity flags are absent; available activity is likewise ignored if sleep is missing. No per-patient reason is emitted. An empty or poorly covered file is a different case: state classification can become UNKNOWN, but the presence of one sleep anywhere switches drift away from its fallback globally (C05).

Targeted repair: Load sources independently and define partially observed states explicitly. Sleep-only information can identify sleep without asserting wake/rest/activity. Record state coverage and fallback reasons separately from file existence.

Acceptance check (not run): All four sleep/steps file-presence combinations plus empty and sparse files; usable information must be retained without inventing the absent state.

7 Design decisions and empirical validation

These 12 findings are not a blanket list of confirmed CPD bugs. They concern intended semantics, model assumptions, conditional upstream limitations or statistical adequacy. For mixed D/R entries, the described source behavior is established, but its scientific acceptability or the appropriate replacement is not established by static inspection. Decide the required behavior or gather independent evidence before changing the algorithm. C25 also contains a narrow source-visible tied-pruning cap discrepancy; that subpoint can be repaired separately without deciding the BOCPD model or gap policy.

7.1 C06: Causal detection cannot repair leaked inputs

Severity: S1, conditional on input. Evidence: D/R.

Location: `wearable/experiment.py`, lines 270–287.

Location: `wearable/steps/detect.py`, lines 374–394.

The default smoothing recipe is `gammadglm`; the existing causality audit documents whole-frame parameter fitting in nine smoothers. Detection accepts those artifacts without a causal-status check. Their early residuals, and histories used to fit whitening/thresholds, can already depend on later observations. This is an upstream issue at the CPD boundary, not evidence that every detector recurrence is noncausal. It particularly affects onset interpretation.

Targeted repair: Record input causal status and calibration interval in manifests; distinguish retrospective comparisons from prospective-valid branches. Use causal initialization or a fixed past calibration for the smoother before certifying the chain.

Acceptance check (not run): Full-chain prefix consistency must include recomputing the upstream transform, rather than truncating a residual computed on a full recording.

7.2 C09: Event-rate calibration admits an always-on detector

Severity: S2. Evidence: D/R.

Location: `wearable/cpd/_whiten.py`, lines 349–370.

Location: `wearable/steps/detect.py`, lines 241–250.

Threshold choice constrains rising-edge counts only. A threshold producing one long alarm throughout 60 worn days can meet a budget even though virtually all readings are flagged. The "at every higher threshold" envelope helps reject low thresholds when intermediate thresholds produce many events, but a flat/degenerate score has no such protection. For a constant finite score, the minimum candidate can be accepted whenever one event fits the budget. Also, each statistic is calibrated in isolation; the final detector and ensemble are not replayed and checked on the null.

Targeted repair: Specify an event lifecycle, occupancy/duration constraint and admissible threshold domain. Record both null event rate and occupancy, and replay final detector/ensemble outputs rather than equating a low onset count with useful alarms.

Acceptance check (not run): Constant scores, sustained plateaus, isolated spikes and correlated streams must not yield near-continuous accepted alarms without an explicit state interpretation. Verify final OR behavior, not just per-stream thresholds.

7.3 C12: Whitening refits leave the latent state in old coordinates

Severity: S2. Evidence: D/R.

Location: `wearable/cpd/_whiten.py`, lines 203–228.

Hourly centers/scales and state corrections are recomputed at each refit, changing the normalized observation coordinate. The OU filter carries its old normalized `x` and `P` into that new coordinate unchanged. A center shift of one old SD does not magically leave the old latent mean valid. New OU amplitude/timescale also changes the interpretation of covariance. Midnight innovations can therefore reflect a parameter-update artifact, not a new physiological change.

Targeted repair: Transform latent mean/covariance into the new normalization where possible, or causally replay a bounded history under the new parameters before publishing new innovations. A reset is another policy but must have explicit settling behavior.

Acceptance check (not run): Force a known normalization change with unchanged raw physiology and compare against a correctly transformed/replayed filter; no artificial spike.

7.4 C13: Kalman drift evidence ignores elapsed time and resets

Severity: S2. Evidence: D/R.

Location: `wearable/cpd/kalman.py`, lines 49–55.

The CUSUM simply adds each finite z; it neither uses elapsed time nor chunk IDs, and does not restart after an alarm. A large accumulated value can persist across hours/days of missingness, then combine with unrelated later evidence. There is no clock-time decay or reset corresponding to the orchestration comment that detectors handle elapsed time. The chosen step changes the evidence per hour. No-reset is an intentional possible monitoring policy, but the gap/duration semantics here are not equivalent to an H8 state flush or independent alarm episodes.

Targeted repair: Decide whether this is cumulative regime evidence or repeated change detection; implement elapsed-time weighting/decay and explicit gap/rearm policy, then recalibrate that same recurrence. Do not reset only at inference.

Acceptance check (not run): Separate two subthreshold excursions by a long gap; check whether their combination is permitted by the chosen policy. Test recovery after a large alarm.

7.5 C16: Drift's analytic threshold does not calibrate its actual statistic

Severity: S2. Evidence: R.

Location: `wearable/cpd/drift.py`, lines 50–59 and 114–132.

The analytic approximation assumes standardized nightly observations of a specified no-change process. The actual u_D uses a selected rolling median/MAD estimated from as few as three nights, a one-device-unit floor, exclusion of alarmed nights and missing-night skips. Its distribution need not be independent $N(0,1)$. One eligible night is not one 24-hour worn day. Consequently assigning this approximation one-fifth of the ensemble's worn-day budget does not verify its rate. The analytic detector is exempt from empirical null-rate reporting.

Targeted repair: Calibrate the entire nightly algorithm on realistic past-only nulls, or state a separate per-eligible-night sensitivity target and validate the approximation.

Acceptance check (not run): Vary baseline length, missing nights, sleep coverage, heavy tails and state availability; measure false verdicts per eligible night and per worn day.

7.6 C18: STS and nightly drift are restricted by transformed availability

Severity: S2. Evidence: D/R.

Location: `wearable/steps/detect.py`, lines 288–308 and 151–154.

All data are first filtered to finite values of the *chosen detector column*. Only then are raw measurements saved for STS and drift. Raw readings in short chunks where a smoother emits NaN, or before a forecast becomes available, are absent. Thus nightly drift, advertised as a raw-channel detector, changes its raw observation support across branches. STS fits and simulates that restricted mask; raw observations that would initialize a real smoother can be missing from its synthetic run. This is more than choosing residual versus raw as a score input.

Targeted repair: Carry the full raw/state stream separately from transformed eligibility. Define which detector needs which support; generate raw STS histories first, then apply the actual transform and eligibility policy once.

Acceptance check (not run): Give two transforms different NaN masks over identical raw observations; raw nightly baselines should agree unless the experiment explicitly varies support.

7.7 C20: Nominal sample counts are presented as actual time coverage

Severity: S3. Evidence: D/R.

Location: `wearable/cpd/_whiten.py`, lines 185–197 and 331–346.

Location: `wearable/cpd/drift.py`, lines 84–85.

Location: `scripts/cpd_calibration_compare.py`, lines 55–64.

Arming uses $n\,\Delta t_{nominal}$, not measured worn duration. One HR observation in a five-minute bin counts five worn minutes after binning; an irregular HRV sample counts ten minutes regardless of actual support. Sleep coverage similarly multiplies reading count by nominal step. Reported exposure on mapped original HR rows instead uses one minute per row, so calibration and reported rates can have different denominators under sparse wear. This proxy may be intentional but is not literal 24-hour wear, and duplicates can inflate it.

Targeted repair: Declare effective-sample exposure versus actual coverage; use one bounded duration/coverage rule consistently for arming, null length, sleep eligibility and reported rates. Never bridge arbitrary gaps by assigning their whole duration.

Acceptance check (not run): Equal sample counts with different occupancy, sparse HR bins and duplicate timestamps; exposure definitions must be documented and consistent.

7.8 C21: Tail fitting and HMM emission scale are inconsistent

Severity: S3. Evidence: D/R.

Location: `wearable/cpd/_whiten.py`, lines 218–224.

Location: `wearable/cpd/hmm.py`, lines 38–47.

The Student-t fit estimates both degree of freedom and scale with location fixed, but stores only degree of freedom. The HMM then assumes scale 1 in state 0 and scale $\sqrt{v_1}$ in state 1. Unit variance and Student-t scale 1 are different: $\mathrm{Var}(t_\nu) = \nu/(\nu-2)$. Robust normalization does not itself prove either scale or variance is one. The initial degree of freedom is then frozen while other noise parameters refit. Threshold calibration may compensate for some mismatch but does not turn posterior state probability into a calibrated physiological risk.

Targeted repair: Choose and fit a consistent emission scale convention; retain fit scale and state whether tails are frozen or refitted. Validate adaptation rather than adding unrestricted full-record HMM training. Rename the stale "Gaussian HMM" label.

Acceptance check (not run): Known t-distributed innovations with multiple scales and tail weights; verify likelihood ratios and calm-state behavior under the chosen convention.

7.9 C22: Baseline exclusion can silently abandon the healthy-history rule

Severity: S3. Evidence: D/R.

Location: `wearable/cpd/_whiten.py`, lines 205–234.

Location: `wearable/steps/detect.py`, lines 224–235.

If abnormal-day removal leaves fewer than ten rows, whitening reintroduces the unfiltered window. If every candidate shuffle date is excluded, calibration uses *all* dates before `p0`, bypassing both exclusion and the nominal trailing window. Slow deterioration can also remain below the abnormal-day

criterion and be absorbed by normal refitting. The final abnormal-date set is passed to calibration without an as-of ledger. Normally past dates are already complete, so this alone is not proof of broad future leakage; initial partial-day classification deserves an explicit boundary check.

Targeted repair: Make fallback choices explicit and recorded; consider freezing the last valid baseline rather than silently treating abnormal data as normal. Supply an as-of exclusion ledger and bounded history to every recalibration.

Acceptance check (not run): Entire trailing window abnormal, fewer than ten healthy rows, persistent subthreshold drift, and arming mid-calendar-day.

7.10 C23: Ensemble severity selects unlike quantities

Severity: S3. Evidence: D/R.

Location: `wearable/steps/detect.py`, lines 94–117.

Location: `wearable/cpd/bocpd.py`, lines 110–112.

Location: `wearable/cpd/hmm.py`, lines 77–83.

Location: `wearable/cpd/kalman.py`, lines 85–91.

Location: `wearable/cpd/kcpd.py`, lines 123–129.

BOCPD/HMM use posterior probabilities; other methods use score divided by twice a threshold. These have no common severity meaning. KCPD thresholds may be negative, making that normalization misleading. Kalman's severity is the maximum over all streams, while phenotype/magnitude use fixed spike-over-ACF-over-drift precedence; its winning severity can describe another stream. NaN severity removes a firing method's phenotype claim while its anomaly still contributes to the OR.

Targeted repair: Preserve all stream scores and select a representative using a declared priority or comparable null-tail measure. Keep winning stream, phenotype and magnitude coupled. Do not interpret primary phenotype as a diagnosed cause.

Acceptance check (not run): Simultaneous stream/detector hits, ties, negative/zero thresholds and nonfinite severities must have deterministic, semantically consistent output.

7.11 C25: BOCPD gap behavior and phenotype are weaker than their labels

Severity: S3. Evidence: D/R.

Location: `wearable/cpd/bocpd.py`, lines 63–94 and 115–125.

The recurrence ignores chunk IDs and increases hazard with elapsed time across a gap. A return reading can therefore favor a new segment even without observed change; settling may suppress reporting but does not clear that evidence. This is an intentional alternative to hard resets, so requires an explicit compatibility decision with H8, not automatic replacement. The new-run prior predictive is a deliberate recursion variant and should be checked against its own stated generative convention, not assumed identical to a textbook implementation.

Phenotype compares two fixed trailing windows rather than the inferred run's segments, without chunk restriction, coverage requirement or uncertainty. Absolute mean and SD differences are heuristically compared; few samples default to a mean shift. Pruning uses a value cutoff, so tied hypotheses can exceed `max_hyp`; the advertised hard computational cap is not strict under ties.

Targeted repair: Specify gap/new-segment semantics; verify a small enumerated Bayesian reference for the chosen change convention. Make phenotype confidence/coverage explicit; enforce a deterministic top-K cap if that cap is required.

Acceptance check (not run): Quiet signal after a long gap, known mean/variance injections, tiny reference sequences and tied hypothesis weights.

7.12 C26: STS fidelity is not established by fitting alone

Severity: S3. Evidence: R, with source-visible approximations.

Location: `wearable/cpd/_sts.py`, lines 99–139 and 155–189.

STS fits a Gaussian structural likelihood, then fits t tails to standardized total forecast errors and applies them only to simulated measurement shocks. Those errors also contain latent-state uncertainty, so their tail weight does not uniquely identify measurement noise. Finite standardized errors are not explicitly masked by observed grid cells before tail fitting; the installed statsmodels treatment of missing-cell residuals must be checked before claiming contamination. No such library behavior was executed/assumed proven here.

Grid collisions keep the last measurement but set exogenous state indicators by any colliding on-state row, potentially pairing an awake measurement with a sleep indicator. Fit output counts observations before deduplication. Simulation keeps timestamps/states but repeats draws anchored at the fitted end state on shifted historical masks; model state/seasonal phase alignment and block independence need library-level verification. Clipping, stationary AR(2) and independent blocks cannot be presumed to reproduce bursts, device pile-ups or slow real variation. These are model adequacy questions even after the coding defects are repaired.

Targeted repair: Jointly align grid values/states with one collision rule; mask tail fitting by observed support; record fit quality and validate the full synthetic distribution, not only fitted coefficients. Verify vendor simulation shock/anchor semantics before claiming exact reproducibility or phase fidelity.

Acceptance check (not run): Colliding samples with conflicting states, long gaps, known seasonal phase, variance/tail checks, burst autocorrelation and fresh versus repeated blocks.

8 Verification and observability gaps

These two findings describe missing assurance and diagnostic information. They do not prove that every affected detector output is wrong. Close them through the specified regression coverage, validation and provenance work, rather than treating them as two additional algorithm failures.

8.1 C30: The present tests leave the central failure modes uncovered

Severity: S2. Evidence: V.

Location: `wearable/selftest.py`, lines 543–578, 580–600 and 627–753.

The main pipeline probe uses one synthetic patient per channel and one prefix cut. Its generated dataframe omits ingest filler rows, so C02 is not exercised. It does not create a first-sleep-in-the-future transition. The HR check excludes the open bin at its cut. Cuts well after a calibration date cannot reveal all future-in-day calibration paths; alarm flags can also remain identical despite changed thresholds. The rate bound permits 2 alarms/day against a 1/3 budget, not target agreement. Checking a threshold's event count on the same null used to choose it is not an independent calibration assessment. The STS coefficient/mask fixture is useful but does not establish real-data synthetic fidelity. Existing full-path probes primarily compare dataframe columns, not a complete as-of fit/calibration dependency ledger.

Targeted repair: Add the acceptance fixtures in this report and independent null/injection validation; test every boundary class rather than one fixed cut. Include output availability, fitted

parameters, threshold histories, feature columns, keys and fallback status. Preserve patient-level statistical validation separately.

Acceptance check (not run): Every D finding has a regression fixture; every R finding has an explicit validation/policy decision. Run these later on an appropriate environment; none were executed during this audit.

8.2 C31: Diagnostic provenance is insufficient for precise reconstruction

Severity: S4. Evidence: D/V.

Location: `wearable/steps/detect.py`, lines 335–361 and 498–517.

The table omits most detector score streams, per-reading thresholds, per-detector availability and the fitted hourly/state/OU parameters. Calibration rows retain thresholds and STS parameters but not every fitted whitening state. An infinite threshold can silently disable an uncalibratable stream without a reason code. The row phenotype "burn-in" also labels settling, and drift's separate minimum-night eligibility is not represented by the shared armed flag. Drift-only runs produce no calibration report because `_thresholds` returns before adding analytic-threshold rows. The incomplete dependency manifest (C04) makes later reconstruction still harder.

Targeted repair: Store compact versioned fit snapshots, validity/reason codes and threshold activation intervals, with optional detailed scores. Always record analytic thresholds and the exposure to which each rate refers. Avoid patient data in public documentation; diagnostic artifacts remain sensitive.

Acceptance check (not run): A known event can be traced from raw support through fit version, threshold, score, eligibility, publication and daily feature without rerunning an unrecorded fitting decision.

9 Cross-cutting decisions before choosing targeted fixes or a rewrite

9.1 What is not itself an implementation bug

Forward-only HMM inference with fixed parameters is valid as a specified model; HMMs do not universally require EM training. BOCPD posterior updating is a fitting operation even without a separate training phase. Patient-local past-only whitening on test patients is permitted by the current pipeline and is not automatically train/test leakage. OR fusion is a recall-oriented choice, not inherently incorrect. STS explicitly rejects forecast surprise/tail rather than pretending to calibrate an unexecuted forecaster. Full-day features support an end-of-day decision, not an alert at the beginning of the same day. Each of these choices still needs accurate claims and suitable validation.

The KCPD MMD expression uses the expected unbiased within/cross-block form; its median kernel bandwidth is permutation-label-invariant within a fixed block. However, exchangeability and the claimed whiteness are assumptions, not consequences of having called a whitening function. The OU-plus-robust-update pipeline does not establish Gaussianity, unit variance or independent tails. The Kalman pair statistic's normal approximation likewise needs enough appropriate innovations; a small coverage threshold is not a proof. These concerns belong in validation, not a claim that the visible formulas are algebraically wrong.

9.2 Minimum repair order

1. Freeze the comparison contract: raw variables, transformed support, availability timestamps, gap/reset semantics, exposure and event definitions.

2. Repair the historical cutoff, observed-row chunking, state fallback and holder boundaries (C01, C02, C05, C07, C11, C19); certify causality only for compatible upstream inputs (C06).

3. Repair feature coverage and dependency keys together (C03, C04). A metadata-only fix is insufficient. Establish a new artifact-version boundary.

4. Repair null construction, missing/degenerate-score handling and repeated-event semantics (C08–C10, C14–C18). Recalibrate after recurrence or eligibility changes.

5. Resolve refit-state coordinates and detector memory policies (C12, C13, C32), then complete exposure, diagnostics, validation and reporting changes (C20–C31).

6. Rebuild affected CPD artifacts and dependent modelling results on the data machine. Compare newly defined results under identical patients and feature bundles; do not silently mix old and new artifact generations.

9.3 When targeted changes are sufficient

If the intended experiment remains patient-local whitening, calibrated detectors and daily classifier features, most D findings are localized to a small number of shared interfaces. Preserve detector mathematics where verified; replace history slicing, chunking, availability mapping, event representation and dependency declarations. Changing all five detectors independently would duplicate repairs and risk changing the experimental question. Statistical R findings still require model validation after the plumbing is fixed.

9.4 When a broader rewrite is justified

A streaming or resumable service needs explicit state and clock events that the current whole-frame API does not expose. A coherent replacement boundary would have:

1. A timestamp-bounded history object containing raw observations, state flags, chunk/coverage intervals, source identity and a known availability time.

2. A fit snapshot recording normalization, OU/tail parameters and its effective time. Refits must causally transform or reconstruct filter state.

3. One null-generation interface returning data plus the same availability, reset and exposure definitions used by real inference.

4. Detector update interfaces with explicit memory/reset policies, score validity and separate state-versus-verdict outputs. Repeated verdicts carry stable event IDs.

5. A publisher that never backdates information, and a daily aggregator that preserves raw variables independently of CPD feature coverage.

6. Complete content-addressed dependencies and diagnostic fit/calibration ledgers.

This is an architectural option, not authorization to change code in this review.

10 Audit coverage and remaining verification

Read in full: all five detector modules, `cpd/__init__.py`, `cpd/_whiten.py`, `cpd/_sts.py`, and `steps/detect.py`. Inspected at their CPD boundaries: ingestion fillers, shared chunking and settings, registry contracts, store keying, CLI construction, experiment recipes, join feature aggregation, calibration reporting, smoke tests and the CPD selftest fixtures. Pipeline documentation and

the CPD implementation history informed interpretation. This is not an audit of every smoother, foundation-model library, classifier or clinical label definition. Their relevant CPD boundary risks are identified above.

No empirical sensitivity/specificity, false-alarm rate, detection delay, computational cost or real-data impact was measured. Static review cannot establish how often the triggering conditions occur. External library numerical behavior and STS fidelity remain explicitly unverified. The acceptance checks define the next investigation without requiring raw data or GPU execution on this laptop during this task.